



Technical Overview:

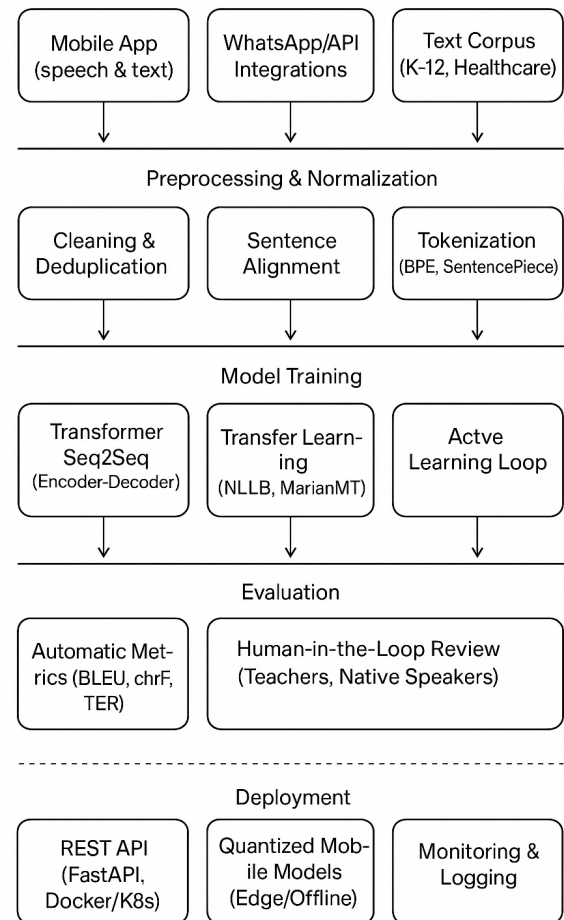
NativeAI

Project Goal:

Bring low-resource African languages online with scalable and cheap AI translation, unlocking access to education, healthcare, and economic opportunities for marginalized communities.

Pipeline Overview

The system is structured as a multi-layer pipeline combining data capture, preprocessing, model training, evaluation, and deployment. This end-to-end design ensures rapid, cost-effective deployment of translation models for spoken-first African languages, even in low-resource settings.



Key Technical Highlights

- 1. Rapid, scalable model deployment**
 - Unified pipeline enables new languages to be launched in weeks.
 - Handles diverse datasets with automatic cleaning, alignment, and tokenization.
- 2. Offline & low-cost optimization**
 - Quantized models run efficiently on entry-level smartphones (<50ms inference).
 - Mobile SDKs and edge deployment make translation accessible in rural areas.
- 3. Adaptive, feedback-driven learning**
 - Active learning loop integrates corrections from teachers and native speakers.
 - Continuous retraining for dialect coverage, improved accuracy, and real-world usability.
- 4. Domain-aware translation**
 - Models can be fine-tuned for critical sectors: education, healthcare, commerce.
 - Enhances trust and reliability in sensitive contexts, as well as cultural alignment.
- 5. Inclusive data acquisition**
 - Gamified apps and messaging integrations encourage speaker participation.
 - Rich, representative datasets improve model robustness and social impact.
- 6. Built on a Proven Foundation: lingo.cm**
 - Leveraging prior experience, where we gathered and processed text corpora
 - Demonstrated ability to design and run cost-efficient inference servers at scale.